Chapter 17. Miller's Planet

If you are familiar with Newton's gravitational laws in mathematical form, then you may find it interesting to explore a modification of them by the astrophysicists Bohdan Paczynski and Paul Wiita (Paczynski and Wiita 1980). In this modification, the gravitational acceleration of a nonspinning black hole is changed from Newton's inverse square law, $g = GM/r^2$ to $g = GM/(r - r_h)^2$. Here M is the hole's mass, r is the radius outside the hole at which the acceleration g is felt, and $r_h = 2GM/c^2$ is the radius of the nonspinning hole's horizon. This modification is a surprisingly good approximation to the gravitational acceleration predicted by general relativity.[60] Using this modified gravity, can you give a quantitative version of Figure 17.2[61] and deduce the radius of the orbit of Miller's planet? Your result will be only roughly correct, because the Paczynski-Wiita description of Gargantua's gravity fails to take account of the dragging of space into a whirling motion by the black hole's spin.

Chapter 25. The Professor's Equation

The meaning of the various mathematical symbols that appear in the Professor's equation (Figure 25.6) is explained on his other fifteen blackboards, which can be found on the web at this book's page at Interstellar.withgoogle.com. His equation expresses an "Action" $\boldsymbol{S}$ (the classical limit of a "quantum effective action") as an integral over "Lagrangian" functions $\mathcal{L}$. These Lagrangians involve the spacetime geometries ("metrics") of the five-dimensional bulk and our four-dimensional brane, and also involve a set of fields that live in the bulk (denoted Q, σ, λ, ξ, and φ^i), and also "standard model fields" that live in our brane (including the electric and magnetic fields). The fields and spacetime metrics are to be varied, seeking an extremum (maximum or